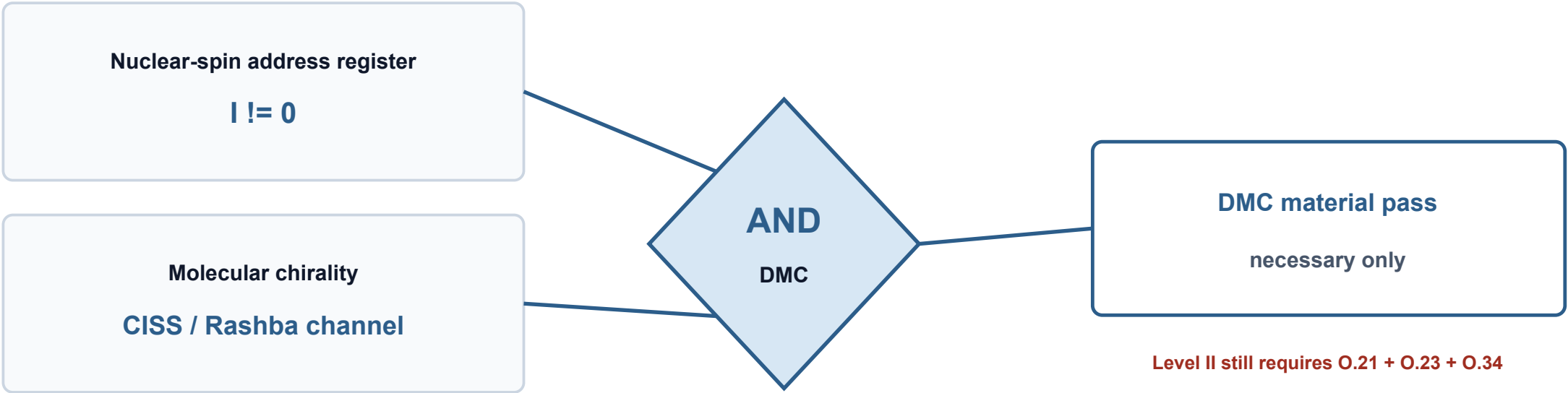


Figure V1.16.2

Dual Material Constraint gate

Level II eligibility requires both a non-zero nuclear-spin address register and molecular chirality.



Substrate	I != 0 leg	chirality leg	DMC gate	engine verdict / caveat
28Si digital control	✗ NO I = 0	✗ NO achiral lattice	FAIL eta_Zeno = 0	TURING_NULL material-DMC failure
natural silicon	M MIXED 29Si low abundance	✗ NO achiral lattice	FAIL missing chiral spin architecture	TURING_NULL by gate no organized CISS channel
beta-tubulin Trp branch	Y YES I != 0 nuclei	Y YES L-chiral protein	PASS material DMC	DMC_PASS_O21_PENDING central eta=0; O.21/O.23/O.34 pending
NV + chiral h-BN cap	Y YES N spin register	Y YES chiral cap	PASS engineered surrogate	DMC_PASS_COOPERATIVE_INVENTORY PRESENT_O21_CLOSURE_PENDING eta_Zeno=26.8; closures pending

Gate discipline: missing either leg sets eta_Zeno = 0; passing DMC is necessary but not sufficient for Level II.